

NETWORK RECONNAISSANCE & PORT SCANNING AUDIT REPORT

Task 03: Network Reconnaissance & Port Scanning Audit

Prepared by: MALLIPUDI JEEVAN KAUSHIK

Assessment date: [Enter actual assessment date]

Tools: Nmap and Wireshark

Operating system: Microsoft Windows

GitHub repository: <https://github.com/jeevankaushik38-code/network-reconnaissance-audit>

Report status: Draft — pending verification of scan results and packet observations

Abstract

This report documents a network reconnaissance and packet analysis exercise conducted as part of Task 03. Nmap was used to perform network reconnaissance and port scanning, and Wireshark was used to capture and inspect network traffic. The reported Wireshark activity included capturing traffic on the loopback interface, attempting to access a local training application at 127.0.0.1:8080, saving a packet capture, and applying the TCP display filter.

The Nmap output files and Wireshark evidence have been reported as uploaded to the associated GitHub repository. However, the exact Nmap results, observed packet details, and verified security findings have not yet been incorporated into this draft. Accordingly, this report does not claim that particular ports are open, specific services are vulnerable, or sensitive information was exposed.

1. Introduction

Network reconnaissance helps identify reachable systems, exposed ports, and services that may require further security assessment. Packet analysis complements this activity by providing visibility into network communications, including protocols, endpoints, and connection behavior.

Nmap and Wireshark serve different but complementary purposes. Nmap provides information about hosts, ports, and services according to the scan performed. Wireshark enables detailed inspection of packets observed on a selected network interface.

This assessment documents the methodology, available evidence, limitations, and recommended reporting practices for an authorized training environment.

2. Objectives

The objectives of this assessment were:

1. Perform network reconnaissance and port scanning against authorized lab targets.
2. Record the ports and services reported by Nmap.
3. Capture and inspect relevant network traffic using Wireshark.
4. Document observed network behavior using screenshots and saved output files.
5. Identify potential security risks only where supported by evidence.
6. Provide recommendations and maintain reproducible assessment documentation.

3. Scope and Authorization

3.1 Assessment scope

The local training application address used during the Wireshark exercise was:

- 127.0.0.1:8080

This is a loopback address that refers to the local computer. The address alone does not establish that the training application was running or that a connection succeeded.

Other Nmap targets must be recorded from the actual scan outputs before the final report is submitted.

3.2 Authorization boundaries

Testing must remain within the explicitly authorized lab scope. Third-party systems, production environments, and other unapproved targets are excluded.

Denial-of-service activity, destructive testing, persistence, and unauthorized data access are prohibited.

3.3 Assessment limitations

The precise assessment window, authorization details, and complete Nmap target list have not yet been verified in this report. These details must be completed using the task instructions and actual assessment records.

4. Tools and Environment

Tool or component	Purpose	Recorded status
Nmap	Network reconnaissance and port scanning	Output files reported as uploaded
Wireshark	Packet capture and inspection	Capture performed
Microsoft Windows	Assessment operating system	Confirmed by assessor

Tool or component	Purpose	Recorded status
Npcap loopback capture adapter	Observe local loopback traffic	Selected for capture
Local training application	Intended local traffic destination	127.0.0.1:8080 attempted

The versions of Nmap and Wireshark, the exact Nmap commands, and the assessment date should be added from the original records if required by the task.

5. Nmap Reconnaissance and Port Scanning

5.1 Objective

Nmap was used to identify network services and port states on the authorized assessment targets.

5.2 Methodology

The Nmap output files were uploaded to the project GitHub repository. The original outputs should be reviewed to extract the actual targets, scan options, port states, detected services, and any version information.

No specific scan command or result is asserted here because the original outputs have not been examined for this report.

5.3 Results

Complete the following table using the saved Nmap results.

```
C:\Users\user>nmap -sT -F 127.0.0.1 -oN tcp-connect-localhost.txt
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-09 17:33 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0012s latency).
Not shown: 96 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
5357/tcp   open  wsdapi
Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds
```

5.4 Interpretation

An open port indicates that a service accepted or responded to a connection attempt under the conditions of the scan. An open port is not, by itself, proof of a vulnerability.

Potential risks should be evaluated using the actual service, its configuration, exposure, software version where reliably identified, and supporting evidence.

5.5 Nmap evidence

```
C:\Users\user>type "%USERPROFILE%\tcp-connect-localhost.txt"
# Nmap 7.991 scan initiated Fri Oct 9 17:41:27 2026 as: nmap -sT -F -oN C:\Users\user\tcp-connect-localhost.txt 127.0.0.1
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00071s latency).
Not shown: 96 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
5357/tcp   open  wsddapi
# Nmap done at Fri Oct 9 17:41:27 2026 -- 1 IP address (1 host up) scanned in 0.33 seconds
```

6. Wireshark Packet Analysis

6.1 Objective

The purpose of the Wireshark exercise was to capture local network traffic and inspect TCP packets associated with the attempted access to the training application.

6.2 Capture methodology

The following actions were reported as completed:

1. Opened Wireshark on Microsoft Windows.
2. Selected the loopback traffic capture adapter.
3. Started a packet capture.
4. Attempted to access `http://127.0.0.1:8080` in a browser.
5. Stopped and saved the capture as `task03-lab-capture.pcapng`.
6. Applied the TCP display filter `tcp`.
7. Saved a screenshot as `03-tcp-filter.png` and uploaded it to GitHub.

These steps describe the reported procedure. The saved capture and screenshot should be inspected to verify the packets and connection behavior.

6.3 Capture details

Parameter	Value
Tool	Wireshark
Operating system	Windows
Interface	Adapter for loopback traffic capture
Intended destination	127.0.0.1:8080
Display filter	tcp

Parameter	Value
Capture filename	task03-lab-capture.pcapng
Screenshot filename	03-tcp-filter.png

6.4 Packet observations

The exact packet observations have not yet been verified. Complete the following fields by reopening the saved capture in Wireshark.

- **Observed protocols:** [Record the protocols shown]
- **Source and destination addresses:** [Record actual values]
- **Source and destination ports:** [Record actual values]
- **TCP connection behavior:** [Record observed handshake, resets, retransmissions, or other relevant events, if present]
- **Application response:** [Record only if supported by the captured packets]
- **Relevant packet numbers:** [Record packet numbers for cited evidence]

The tcp display filter shows packets Wireshark identifies as TCP. It does not, by itself, demonstrate that a connection succeeded or that a vulnerability exists.

6.5 Wireshark evidence

task03-lab-capture.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	56	50335 → 8080 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256
2	0.000018700	127.0.0.1	127.0.0.1	TCP	44	8080 → 50335 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
3	0.019321600	127.0.0.1	127.0.0.1	TCP	56	50336 → 8080 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256
4	0.019342600	127.0.0.1	127.0.0.1	TCP	44	8080 → 50336 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Frame 1: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 50335, Dst Port: 8080, Seq: 0, Win: 65535, Len: 0

0000 02 00 00 00 45 00 00 34 a2 d7 40 00 80 06 00 00E..4..

0010 7f 00 00 01 7f 00 00 01 c4 9f 1f 90 40 e0 5b 91:..

0020 00 00 00 00 80 02 ff ff f6 48 00 00 02 04 ff d7:..

0030 01 03 03 08 01 01 04 02:..

Packets: 4 · Displayed: 4 (100.0%) · Dropped: 0 (0.0%) | Profile: Default

7. Correlation of Nmap and Wireshark Results

Nmap and Wireshark provide different forms of evidence. Nmap reports port and service information under the conditions of its scan, whereas Wireshark shows traffic actually observed on the selected interface during the capture period.

Assessment question	Evidence required	Current status
Which ports were open?	Nmap scan output	Results not yet transcribed
Which services were detected?	Nmap service results	Results not yet transcribed
Which TCP packets were observed?	Wireshark capture	Capture reported as saved; packet details pending verification
Did the local application respond?	Browser result and/or packet evidence	Not established by this report
Was a security vulnerability confirmed?	Reproducible supporting evidence	No confirmed finding documented

A difference between Nmap and Wireshark observations does not automatically indicate a security issue. Differences may arise from interface selection, capture timing, network configuration, or scan conditions.

8. Risk Assessment

Potential security risks must be evaluated using verified findings rather than assumptions.

Finding ID	Observation	Potential impact	Severity	Recommended action
F-01	[Insert verified observation]	[Evidence-based impact]	[Justified rating]	[Appropriate mitigation]
F-02	[Insert verified observation, if applicable]	[Evidence-based impact]	[Justified rating]	[Appropriate mitigation]

No confirmed vulnerability or severity rating is assigned in this draft because the actual scan and packet findings have not yet been reviewed.

8.1 Risk evaluation criteria

- **Low:** Limited potential impact and restricted exposure.
- **Medium:** Meaningful impact requiring mitigation.
- **High:** Significant potential impact requiring priority attention.

These are qualitative descriptions. Any final severity rating should be justified by the observed evidence, likelihood, and potential impact.

9. Recommendations

The following recommendations apply to the authorized lab environment:

1. Verify all assessed targets against the written authorization and approved inventory.
2. Review exposed services and disable unnecessary services where appropriate.
3. Restrict service access to authorized hosts and networks.
4. Apply relevant security updates and configuration hardening.
5. Keep intentionally vulnerable lab systems isolated from production networks and the public internet.
6. Use Wireshark captures to support findings with actual packet evidence.
7. Store evidence securely and redact credentials, session cookies, and other sensitive data.
8. Repeat testing after remediation only when authorized.

These are general recommendations, not claims that a particular weakness was found during this assessment.

10. Conclusion

This assessment documents the reported use of Nmap for network reconnaissance and Wireshark for packet capture in an authorized training context. The Wireshark workflow included selecting the loopback interface, attempting to access a local training application, saving a packet capture, and applying a TCP display filter.

The Nmap outputs and Wireshark evidence were reported as uploaded to the project repository. However, the exact port states, detected services, packet observations, and potential security findings remain to be verified from those artifacts.

The final assessment conclusions should be completed after the evidence has been reviewed and the results documented accurately. No vulnerability should be claimed without supporting evidence.

11. Evidence Appendix

GitHub repository:

<https://github.com/jeevankaushik38-code/network-reconnaissance-audit>

Evidence expected in the repository:

1. Original Nmap scan output files.
2. 03-tcp-filter.png.
3. task03-lab-capture.pcapng, if retained and confirmed safe to publish.
4. This completed Network Audit Report PDF.
5. Any additional evidence screenshots required by the task instructions.

Final verification checklist

- Assessment date and any required authorization details completed.
- Actual Nmap targets, port states, and services entered.
- Actual Wireshark packet observations entered.
- Evidence screenshots inserted and captioned.
- Potential findings and severity ratings supported by evidence.
- Sensitive data removed from public artifacts.
- Final PDF uploaded to the GitHub repository.

End of report